

Bayesian Inference of the Higgs Boson Mass from CMS Four-Lepton Data Using MCMC Methods

Oren Levis, s5504945
Irena Garbasi, s5488710
Ege Acar Eti, s5515645
Barnabás Egyed, s5638550
Madelief Jeuken, s5746817

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Abstract

The Higgs boson is a short-lived particle in the standard model that cannot be observed directly, but can be identified through its decay products[1]. This project aims to reconstruct its mass from the invariant mass of the four-lepton final states. Raw data from the CMS experiment at CERN were filtered to extract only relevant events, and the invariant mass of the four-lepton system was computed for four datasets. The data were modeled as a combination of a Gaussian signal and exponential background, and a Poisson likelihood was used to compare model predictions with observed counts. The posterior distribution of the model parameters was computed using Markov Chain Monte Carlo (MCMC) sampling. The Higgs mass found was $m_H = 123.8^{+2.3}_{-1.5}$ GeV. This value is consistent with the experimental results[1], demonstrating that this method provides a reliable estimate of the Higgs boson properties.

Contents

1	Introduction	3
2	Physical Background	3
3	Data	4
4	Model	5
5	Statistical Method	6
5.1	Likelihood	6
5.2	Priors	6
6	Inference Procedure	7
7	Results	8
7.1	Model Comparison With Data	9
8	Discussion	10
9	Possible Extensions	10
10	Conclusion	10

A Additional Figures	13
B Reproducibility	14
C Feedback	14

1 Introduction

The Standard Model (SM) of physics has provided multiple highly accurate predictions of (non)collider based experiments throughout the years, and has been tested to a surprisingly high extent [2–5]. It describes (anti)matter through fermions, i.e. quarks and leptons, and their interactions via gauge bosons: the photon γ for the electromagnetic interaction, the W and Z for the weak interaction, and the gluons g for the strong interaction.

For a long time, it remained unclear how the gauge bosons obtained their rather large masses, until it was postulated that a new complex scalar doublet can lead to their generation. Often referred to as the "last piece to the SM", this new boson could further explain the fermion masses through Yukawa couplings [1]. This particle cannot be observed directly, yet it can be identified through its interactions; as with any particle physics experiment, searches comprise of predicting possible decay and production modes, and examining the invariant masses of the products involved. Background events are filtered through various selection criteria, and the suspected new particle appears as a statistically significant peak above a background-only model. The goal of this project is to reconstruct this peak for the publicly available CERN decay data [6], and obtain the model parameters (including the Higgs mass) through a Bayesian inference framework.

This report is structured as follows: First, the physical background behind the measurement of the Higgs boson is explained, and the relevant experimental measurements are outlined in section 2. Next, the data extraction methods are discussed in section 3, along with the selection filters applied to obtain the candidate Higgs decays. In sections 4, 5, 6 we outline the model used to describe the data, statistical methods and choices of priors, along with the computational procedure to infer the model parameters. We present the final parameter estimates and their correlations in section 7, and end with a discussion and suggestions for possible extensions.

2 Physical Background

It has been shown that a symmetry breaking mechanism on the electroweak theory via the introduction of a complex scalar field leads to the generation of the W and Z masses, and to an additional scalar elementary particle coined the Higgs Boson (H) [7–9]. Though accurately predicted, its mass m_H remains a free parameter in the SM, and must therefore be experimentally constrained [1]. Various experimental searches have been conducted to directly observe Higgs bosons, and recent measurements have limited m_H to within the range 114.4 – 152 GeV at 95% Confidence Level (CL). [10, 11]

There exist various prominent decay channels from which m_H can be inferred. A typical example of a sensitive decay mode is $H \rightarrow 4\ell$, where ℓ stands for any lepton. Imposing charge and lepton flavour conservation leads to three main decay channels:

$$\begin{aligned} H &\rightarrow ZZ \rightarrow 2e^-2e^+ \\ H &\rightarrow ZZ \rightarrow 2\mu^-2\mu^+ \\ H &\rightarrow ZZ \rightarrow 2\mu^\pm 2e^\pm \end{aligned}$$

Therefore, from an analysis of the distribution of the 4-lepton invariant masses, one can infer m_H . The resulting histogrammed data is expected to have two significant peaks, one at 91 GeV (i.e. the Z resonance) and one at m_H . From momentum measurements, one can calculate the invariant mass of the leptons through

$$E^2 = p_x^2 + p_y^2 + p_z^2 + m^2 \tag{1}$$

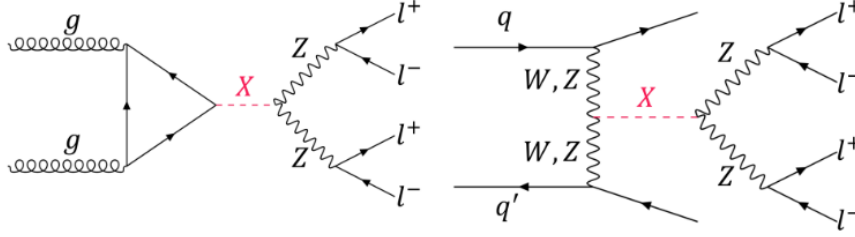


Figure 1: Feynmann diagrams showing the production and decay of heavy Higgs bosons X . Left: gluons $\rightarrow X \rightarrow ZZ \rightarrow 4\ell$. Right: vector bosons $\rightarrow X \rightarrow ZZ \rightarrow 4\ell$. Adapted from [12]

where m is the measured lepton mass, and p_i are the momentum components in each direction. Though detectors do not measure the components directly, their magnitudes can be inferred through the following relations, where η is the lepton pseudorapidity, ϕ is its azimuth angle and p_T its transverse momentum [13]:

$$p_x = p_T \cos \phi \quad p_y = p_T \sin \phi \quad p_z = p_T \sinh \eta \quad (2)$$

3 Data

The raw data was recorded in 2012 by the CMS experiment at CERN[6]. It consists of electron and muon signals received by the detectors. It is in NanoAOD format which is compatible to be used with the uproot package in Python directly unlike regular AOD files. The files have a single tree called 'Events' which was accessed first in chunks to not use a lot of RAM simultaneously as the data is large.

Afterwards, the relevant data was collected in an awkward array for several filters to be applied for the event selection. These filters were obtained from the filtering code in the related Github repository of the data[14] and the discovery paper from CERN[1]. The selected events had the following properties: Transverse momentum p_T below 5 and 7 GeV for muons and electrons respectively, p_η below 2.4 and 2.5 for muons and electrons so that they are closer to the central axis, pfRelIso below 0.4 so that the decay products are isolated, the impact parameters $|d_{xy}| < 0.5$ and $d_z < 1$ and the significance $SIP_{3D} = \sqrt{\frac{d_{xy}^2 + d_z^2}{\Delta d_{xy}^2 + \Delta d_z^2}}$ below 4 to make sure the particles came from a Z or Higgs decay.

Next only the events that result in 4 leptons being formed are selected and divided into $4e$, $2e2\mu$ and 4μ channels so that the mixed channel events are only counted once from one dataset. Also it is checked that the total charge of the 4 leptons adds up to zero. Then the leptons in each event are sorted from highest p_T to lowest. Events with the highest and the second highest lepton having $p_T > 20$ GeV and $p_T > 10$ GeV respectively were selected to check if the decay products are energetic enough to have come from a Higgs decay.

The final filter calculates the mass of the Z bosons that the lepton pairs could have decayed from. The closest mass to the actual Z boson mass is named m_{Z_1} and the other one is named m_{Z_2} . Events that have $40 < m_{Z_1} < 120$ and $10 < m_{Z_2} < 120$ were selected so that at least one Z boson is on-shell and that the Z bosons had enough energy to have come from a Higgs decay. Then the invariant mass for the final events were calculated. This was done for 4 datasets and then they were combined and exported as a .csv file to be used in the next stages.

4 Model

The invariant mass distribution of four-lepton events, denoted by $x = m_{4\ell}$, is modelled as the sum of a signal component and a background component:

$$m(\theta) = S(\theta_s) + B(\theta_b), \quad (3)$$

where θ represents the full set of model parameters. The signal component, corresponding to the Higgs boson peak, is modelled using a Gaussian distribution:

$$S(x|A, m_H, \sigma) = A \exp\left(-\frac{(x - m_H)^2}{2\sigma^2}\right),$$

where $\theta_s = (A, m_H, \sigma)$. The background is described using an exponential function:

$$B(x|B_0, k) = B_0 \exp(-kx).$$

where $\theta_b = (B_0, k)$. Therefore, the full model prediction is:

$$m_i(x|\theta) = A \exp\left(-\frac{(x - m_H)^2}{2\sigma^2}\right) + B_0 \exp(-kx).$$

where $\theta = (m_H, \sigma, A, B_0, k)$. Given a set of bin centres x_i , the model predicts the expected number of events in each bin as:

$$m_i(x_i|\theta) = A \exp\left(-\frac{(x_i - m_H)^2}{2\sigma^2}\right) + B_0 \exp(-kx_i). \quad (4)$$

The parameter vector is given by:

$$\theta = (m_H, \sigma, A, B_0, k). \quad (5)$$

Each parameter has a clear visible interpretation in the model:

- m_H : The Higgs boson mass corresponds to the position of the signal peak.
- σ : The detector resolution determines the width of the Gaussian peak.
- A : The signal amplitude is proportional to the number of Higgs events observed.
- B_0 : The normalization of the background contribution.
- k : The exponential decay constant controls the slope of the background.

There are three main features that the model is designed to present from the data. First, the model is steered to display the localised peak near $m_{4\ell} \approx 125$ GeV. This is where the Higgs boson signal is expected to be according to literature. This localised peak is modelled using a narrow Gaussian plot due to the rise in events around 125 GeV.

Secondly, the data contains contributions from non-resonant Standard Model processes, such as continuum ZZ^* production. These processes produce a broad distribution of invariant masses without sharp features. The background is modelled using an exponential function, which captures the approximately monotonic decrease in event rate with increasing mass in the selected range. This is dependent on the parameters B_0 and k .

Lastly, the model needs to capture the relative contribution of signal and background across the mass range. At masses far from the Higgs peak, the distribution is dominated by background

events, while near $m_{4\ell} \approx 125$ GeV, the signal produces a statistically significant excess above the background expectation. The relative magnitudes of A and B_0 determine how prominent the signal appears compared to the background.

However, the model also includes some simplifications that must be considered. For one, the signal and the background are modelled using a Gaussian and exponential plot respectively, which both introduce assumptions. The Gaussian plot assumes that the energy measurements are symmetric, while the exponential plot assumes a smooth and monotonically decreasing distribution of events, when in reality more complex processes could be taking place. Furthermore, the data is binned to decrease computation time, which results in a loss of information of the distribution of events *within* the bins.

There are also uncertainties that arise from the Poisson distribution assumption as well as systematic assumptions that are not included. By assuming the events in each bin follow an independent Poisson distribution, any potential correlation in fluctuations between bins are ignored. There are also systematic errors that come from the detector that are not included, despite potentially leading to an understated total uncertainty of the parameters. These justified simplifications allow for reasonable but simple descriptions of the signals and parameters, resulting in a clear interpretation of the data.

5 Statistical Method

The data analysis was performed within a Bayesian inference framework, where the probability distribution of the model parameters is determined given the observed data. The model predictions are compared to the observed data through a likelihood function, which quantifies how well the predicted number of events matches the measured counts in each bin of the mass distribution $m_{4\ell}$. Therefore, the comparison between the data and the model is done on a bin by bin basis, using observed and predicted counts.

5.1 Likelihood

Since the data consisted of event counts per bin, a Poisson likelihood was used. The likelihood represents the probability of observing a given number of events in each bin, given the expected number the model predicts.

The log-likelihood was used for computational purposes:

$$\ln \mathcal{L}(\theta) = \sum_i [y_i \ln m_i(\theta) - m_i(\theta)] \quad (6)$$

where

- y_i is the observed number of events in bin i
- $m_i(\theta)$ is the model prediction

Terms such as $\ln(y_i!)$ have been omitted from the likelihood function, as they do not depend on the model parameters, therefore they do not affect the inference. Furthermore, it was assumed that the uncertainties follow Poisson statistics and that the counts in different bins are statistically independent.

5.2 Priors

The model used independent uniform priors with hard bounds instead of Gaussian priors centred around a guessed value. The ranges were chosen so that they make physical sense, but

were also tested and refined through trial and error using pre-MCMC diagnostic such as testing the likelihood function and the mean acceptance rate for the walkers. These initial evaluations were only used to identify regions that produced an unstable or non-physical model.

The priors for the Higgs mass were set from 120-130 GeV since the Higgs mass is expected to appear around 125 GeV. The width parameter σ was restricted to a value of 1-2 GeV because the width of the reconstructed peak is dominated by the detector resolution instead of the Higgs width. The signal Strength A (0-5000) and background normalization B_0 (0-100000) were both restricted to positive values because event counts cannot be negative. The upper bound was set high enough for A to allow for a visible signal without unrealistic high peaks and since B_0 dominates the counts from the data, a broader range was used and tested. The background k (0-0.2) is modelled as a falling exponential so restricting it to a positive value enforces the correct physical shape, and the upper bound was again chosen with the help of the pre-MCMC diagnostic tests to be broad enough so that it doesn't bias the results while still producing visible and clear results.

6 Inference Procedure

The posterior distribution of the parameters was obtained using Markov Chain Monte Carlo (MCMC) sampling. This was done using the emcee library, which implements an ensemble sampler, based on multiple parallel chains called walkers. A total of 30 walkers were used to explore the parameter space, which consisted of the 5 parameters in the model. Each walker was evolved for 5000 steps, resulting in 150,000 initial samples. However, the first 1000 steps are discarded as burn-in, leaving 120,000 posterior samples used for inference. This assures that the remaining chains are not influenced by their starting positions.

The convergence of the MCMC chains were assessed using trace plots for all 5 of the parameters (Figure 4). The walkers for mass and width parameter can be seen moving around the high probability region creating a fuzzy graph which means that the walkers are exploring every place uniformly and do not produce any long term patterns. A , B_0 and k all converge and settle into a clear plateau showing that the model discards the learning data before the burn in cut-off. The reason the Higgs mass and width parameter exhibit a broader fluctuation is due to their larger statistical uncertainty.

The burn in region was discarded before forming the final posterior samples which were then summerized using a corner plot. This corner plot shows the marginal distribution of each parameter together with the correlations between these parameters. The median of the posterior samples were used to estimate each parameter and the 16th and 84th precentile were used to define the central 68 precent credible interval. Taking into account the asymmetry of error this provides a Bayesian estimate of the parameter uncertainties in the form of:

$$\theta = \theta_{50}^{+(\theta_{84}-\theta_{50})}_{-(\theta_{50}-\theta_{16})} \quad (7)$$

7 Results

Figure 2 shows the posterior distributions and correlations between parameters obtained using the MCMC algorithm. The final estimates are reported with their confidence intervals in Table 1. It is important to note that the results agree with the currently accepted value of m_H [1], yet the width of the Higgs spectrum is not strongly fixed by the data: the broad posterior for σ suggests that more data points or higher detector resolutions would significantly improve the results of this analysis.

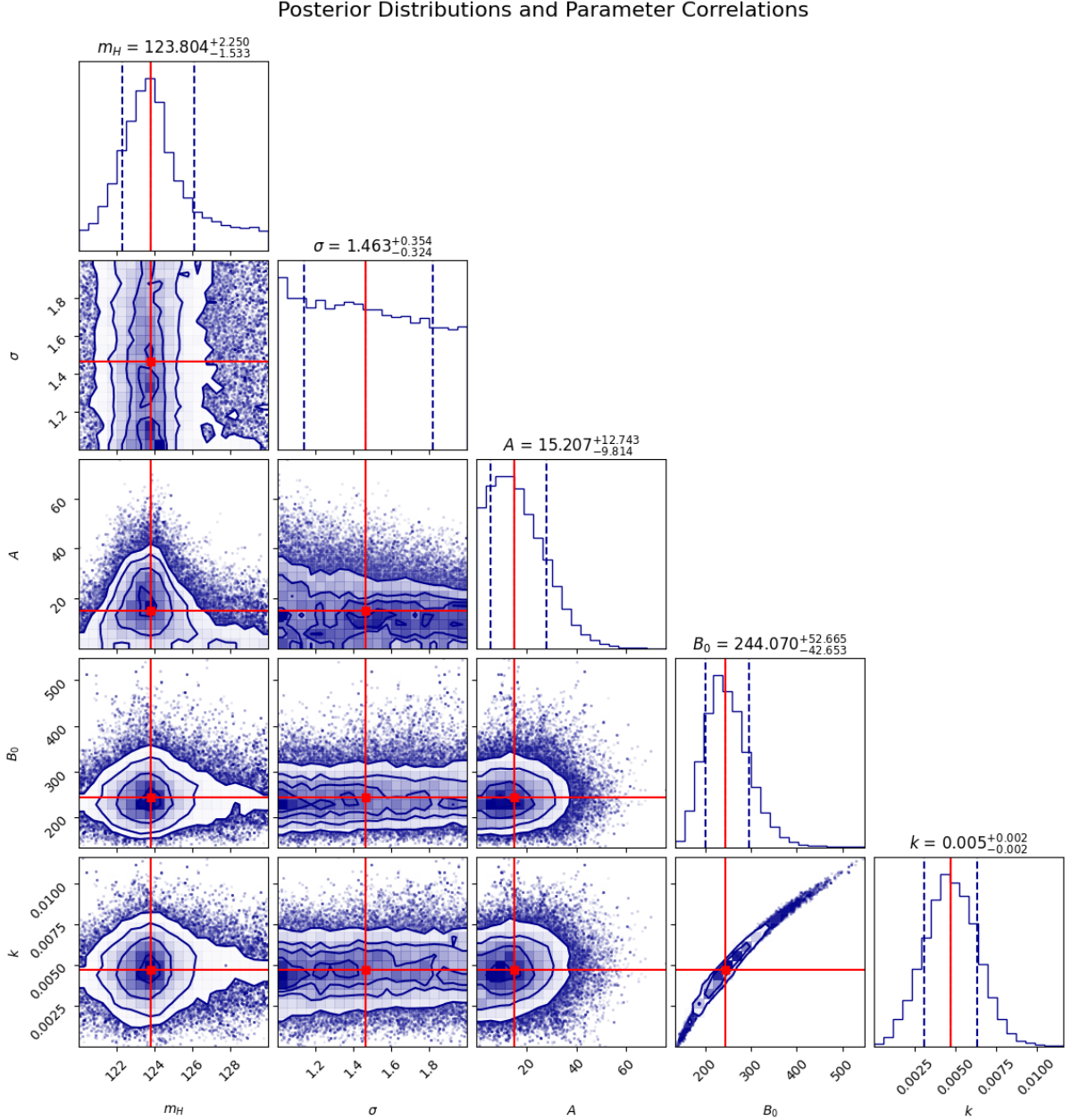


Figure 2: Corner plot for the relevant model parameters.

Parameter	Estimate	1-sigma uncertainty
m_H	123.8	+2.3 -1.5
σ	1.5	+0.4 -0.3
A	15.2	+12.7 -9.8
B_0	244.1	+52.6 -42.6
k	0.005	+0.002 -0.002

Table 1: Posterior parameter estimates. Note the extra decimal places needed for k

7.1 Model Comparison With Data

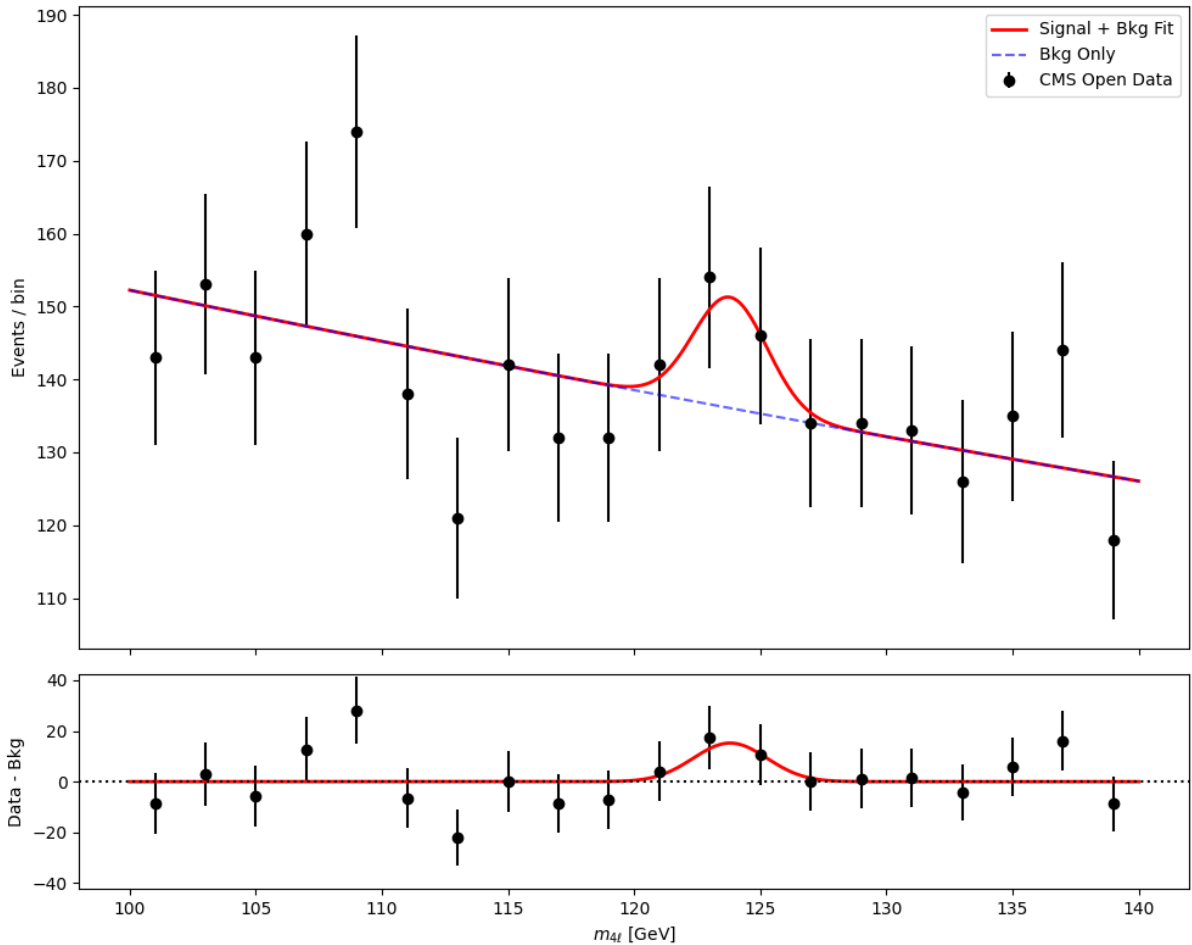


Figure 3: Comparison between data and model prediction. The top panel shows the full spectrum with and without background, and the bottom panel shows the residuals.

The fit and residuals shown in Figure 3 indicates an agreement between the data and the model including the massive Higgs boson; It is shown that a background-only model is not sufficient to explain the large peak around the suspected Higgs mass of 125 GeV. Note the large Poisson errors arising from the small number of data points ($\sim 1/\sqrt{N}$).

It should be noted that the majority of the posterior distributions shown are somewhat asymmetric, further motivating the use of quantiles rather than the standard deviation. In the case of the parameter k , the value of 0 is well within the reported CL, suggesting that the model cannot distinguish between a flat ($k = 0$) background, or a slowly varying exponential decay.

Nonetheless, it is clear that despite the shape of the background, the amplitude of the signal A has a distribution far from the null value; this further supports the fact that a background-only model cannot explain the experimental data.

8 Discussion

The result obtained for the Higgs mass does not exactly agree with the literature value but it is still covered in the 1σ credible region in the posterior[1]. The mass spectrum with the best fit plot describes the data well as 4 out of 20 error bars are left out which is less than the 32% expected ratio. Compared to the plot from CMS there are less data points and larger error bars. This points to the limitation that the data used in this project was more limited compared to the CMS experiment. Also since the NanoAOD files were used, a few of the keys of the events were not accessible so less filters could be used in the event selection, which could have led to more background data in Figure 5.

The σ parameter's posterior shows that no value is strongly preferred over another, so the prior could be broadened. However, the result found is consistent with the experimental resolution for CMS[1]. The results for the A and B_0 parameters are also consistent with experiments as roughly 10-50 Higgs signals are expected to be observed on top of hundreds of background signals[1]. The results for the k parameter show that the background is more linear than exponential compared to experimental results and it is also highly correlated with B_0 . This could be due to the limitation with filters mentioned above or it could show that other background models could fit well too. Another limitation is that the binning of the data influences the results as well. A bin number of 50 was chosen as a lower one deleted many details and a higher one made more noise apparent.

9 Possible Extensions

Due to the result for k , a linear background model could be used to fit the data as well to see the results. To decrease the correlation between k and B_0 the sample chain could be thinned. A model comparison can be performed through nested sampling comparing the only background model to the signal+background one. Different values for the priors or different priors altogether could be tested. However, the main improvement would come from incorporating more data from experiments at CERN.

10 Conclusion

To conclude, the mass of the Higgs boson was determined using the invariant mass of the four lepton final states from filtered data of the CMS experiment at CERN. The four-lepton mass distribution was modelled using a signal-plus-background model. The signal follows a Gaussian distribution, while the background consists of an exponential function. The data was analyzed within a Bayesian inference framework. After defining a Poisson likelihood function and independent uniform priors with hard set bounds, MCMC was used to sample from the posterior to estimate parameters and uncertainties. The mass spectrum was then plotted for a visual comparison between the data and the model prediction. Corner plots were also created for a better visualization of the model parameters.

The final result was $m_H = 123.8^{+2.3}_{-1.5}$ GeV, which falls within the 1σ credible region of the experimental value of the Higgs mass[1]. The results for the other parameters were also found to be consistent with experimental values, with the exception of parameter k , suggesting the

background is more linear than exponential.

These results are significant, as they indicate that this method is reliable in determining Higgs boson properties.

Author Contributions

- Ege: data processing, event selection, wrote the Data, Discussion and Possible Extensions sections
- Oren: data processing, repository integration, code cleanup, wrote Introduction, Physical background, Results and Appendices
- Irena: statistical inference, wrote Abstract, Statistical Method(until 5.2), part of Inference Procedure, and Conclusion
- Barnabás: choosing correct priors, convergence analysis, posterior and final spectrum visualisation, wrote Priors (5.2) and part of Inference Procedure (6)
- Madelif: created model and wrote Model section of report

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A Additional Figures

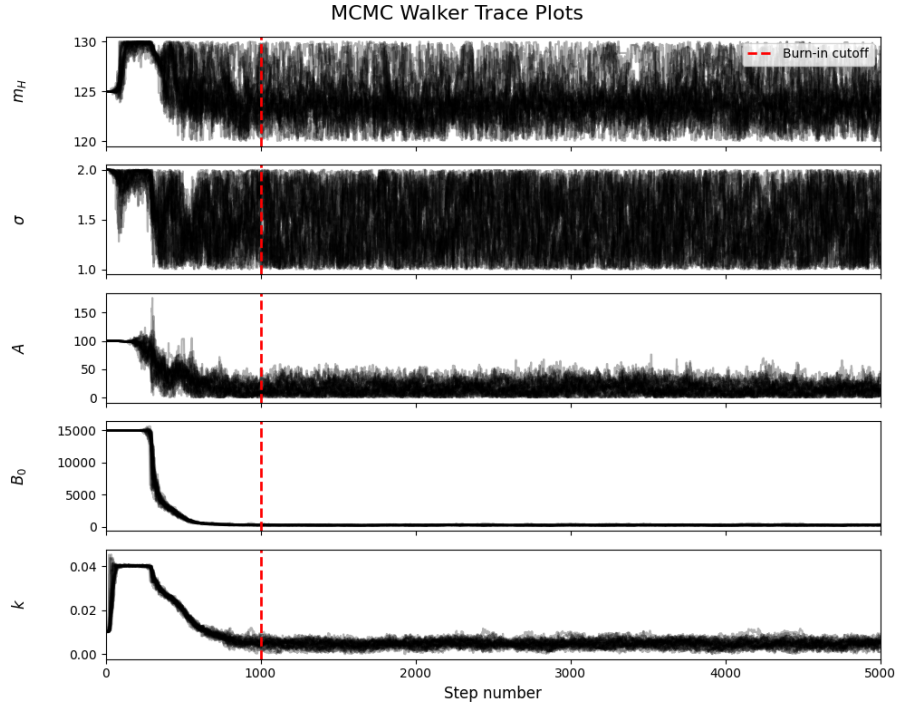


Figure 4: Trace plot from the MCMC walkers for each parameter, including the burn-in cutoff point.

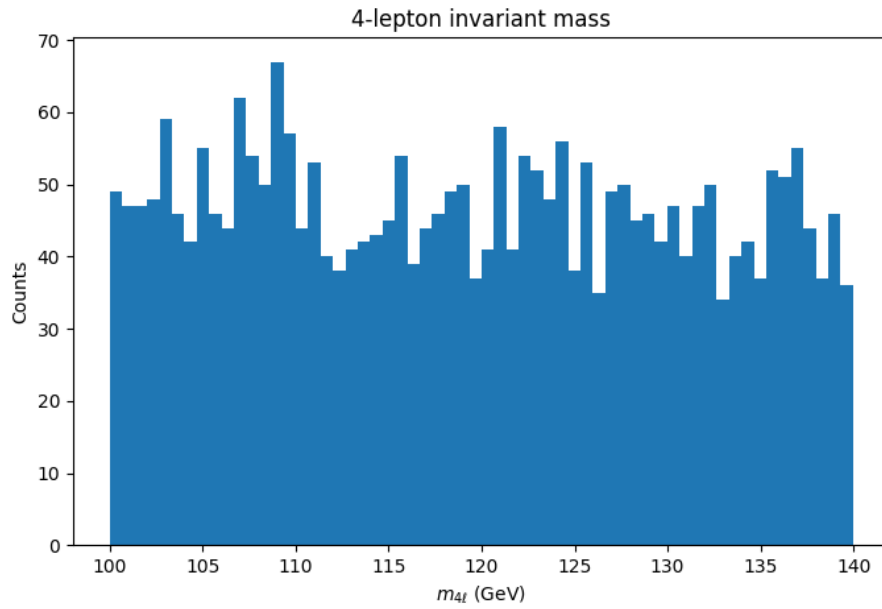


Figure 5: Histogram of the invariant masses $m_{4\ell}$ after applying the selection criteria.

B Reproducibility

Open-source CERN data in the form of nanoAOD files were used to reproduce the discovery plot of the Higgs boson from 4 leptons [6]. These files were processed using Python, with important packages including `uproot`, `pandas`, `awkward` and `emcee`, `corner` for the analysis of the data.

The notebooks used to produce the results shown above can be found in [this](#) GitHub repository under `notebooks`. All information required to reproduce the analysis is contained in the README file, and in the text inside the notebooks.

C Feedback

Please provide any feedback on the project, including what you found most challenging and any suggestions for improvement. How doable was this project scale 1-10.

Barnabás: Moderately challenging but doable in time - 6.5/10 (10 being the easiest)

Ege: The data in the link given in the project Overleaf (AOD format) was not compatible with uproot and we did not feel comfortable working with C++ and ROOT, so we found NanoAOD files of CMS data ourselves which worked with uproot[6]. Then figuring out all the necessary filters was fun but since there is a lot of data, a lot of time was spent waiting for the code to run in previous iterations of the code. - 7/10